

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes

Annexure A

Data Interpretation

Code of Conduct:

I M. Pranay (192525382)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Pranay

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Structure from Motion Consistency

Two datasets produce:

- Dataset A: accurate depth but requires high-quality images
- Dataset B: inconsistent depth but works on low-quality images

Compare the datasets and evaluate which setup is more practical for real-world deployment. Justify your reasoning.

A.

Dataset A provides **accurate depth estimation**, but it requires high-quality images. Dataset B works with low-quality images, but its depth estimation is inconsistent.

Comparison:

- **Dataset A**
 - Produces accurate and reliable depth information.
 - Requires high-quality images.
 - More suitable when good cameras and proper lighting are available.
 - Provides better results for applications requiring precise 3D reconstruction.
- **Dataset B**
 - Can operate with low-quality images.
 - More flexible in poor image conditions.
 - Produces inconsistent depth, which can reduce reliability.
 - May be useful when image quality or camera quality is limited.

Evaluation:

For real-world deployment, **Dataset A is generally more practical when accuracy and reliability are important**. Although it requires high-quality images, its accurate depth estimation provides more consistent results.

Dataset B is preferable when the system must operate with low-quality images and accuracy requirements are less strict.

Conclusion:

Dataset A should be selected for applications such as **3D reconstruction, robotics, and depth-based object analysis**, where reliable depth information is essential. Dataset B can be selected for resource-constrained environments where image quality cannot be controlled.

2. Stereo Vision vs Motion Estimation

Depth estimation results:

- Stereo Vision: accurate but requires dual cameras
- Motion Estimation: less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

Answer:

Stereo Vision and Motion Estimation are both techniques for estimating depth, but they have different hardware and accuracy requirements.

Comparison:

- **Stereo Vision**
 - Uses two cameras.
 - Provides more accurate depth estimation.
 - Requires additional hardware.
 - Camera calibration and synchronization are required.
 - Suitable when hardware resources are available.

- **Motion Estimation**
 - Uses a single camera.
 - Requires less hardware.
 - Generally provides less accurate depth estimation.
 - More suitable for low-cost and compact systems.
 - Useful when only one camera is available.

Evaluation under Hardware Constraints:

When hardware availability is limited, **Motion Estimation is preferable** because it requires only a single camera. It reduces hardware cost and system complexity.

However, if accurate depth estimation is more important than hardware cost, Stereo Vision is the better choice.

Conclusion:

For **hardware-constrained applications**, Motion Estimation is preferable because of its single-camera requirement. For applications requiring **high depth accuracy**, Stereo Vision should be selected despite its additional hardware requirements.

3. Model Generalization vs Overfitting

Performance:

Model Training Testing

A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

Answer:

The two models have the following performance:

Model Training Accuracy Testing Accuracy

Model A	98%	70%
Model B	90%	88%

Model A:

- Training accuracy is very high at **98%**.
- Testing accuracy decreases significantly to **70%**.
- There is a large gap of **28 percentage points**.
- This indicates that Model A is likely **overfitting** the training data.
- Its performance on unseen data is less reliable.

Model B:

- Training accuracy is **90%**.
- Testing accuracy is **88%**.
- The difference is only **2 percentage points**.
- This indicates better generalization.
- It is more likely to perform consistently on unseen data.

Evaluation:

Model B is more reliable for deployment because it maintains high performance on both training and testing data. Model A may perform very well on known training data but could fail when exposed to new real-world inputs.

Conclusion:

Model B should be selected for deployment. Its smaller training-testing gap demonstrates better generalization and lower overfitting risk. Although Model A has higher training accuracy, its low testing accuracy makes it less suitable for real-world deployment.

4. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

Answer:

The YOLO detection results are:

Object Scale Detection Accuracy

Large	95%
Medium	85%
Small	60%

Comparison:

- YOLO performs best on **large objects**, achieving **95% accuracy**.
- Performance decreases to **85%** for medium-sized objects.
- Performance is significantly lower for **small objects**, at only **60%**.
- This shows that object scale has a strong effect on detection performance.

Evaluation:

YOLO alone may not be sufficient for multi-scale detection tasks because its performance drops considerably for small objects. Small objects contain fewer pixels and therefore have less visual information available for detection.

Possible Improvements:

- Use **multi-scale feature extraction**.
- Apply feature pyramid or multi-level feature techniques.
- Increase input image resolution.
- Use data augmentation containing different object sizes.
- Improve training with more small-object examples.
- Use specialized detection techniques for small objects.
- Optimize anchor settings or detection heads where applicable.

Conclusion:

YOLO provides excellent performance for large objects and good performance for medium objects, but its **60% accuracy for small objects indicates a limitation**. Therefore, YOLO alone may not be sufficient for reliable multi-scale detection. Additional multi-scale features, higher-resolution inputs, and suitable training strategies should be used to improve small-object detection.

5. Motion Model Selection for Animation

Two models:

- Model A: smooth but physically inaccurate
- Model B: realistic but computationally intensive

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

Answer:

The two motion models have different advantages:

- **Model A**
 - Produces smooth motion.
 - Computationally efficient.
 - May run effectively in real time.
 - Physically inaccurate.
 - Suitable when visual smoothness and speed are more important.
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- **Model B**

- Produces realistic and physically accurate motion.
- Requires more computational resources.
- May be slower to execute.
- Suitable for applications where realism is the primary requirement.

For Real-Time Animation:

Model A should be preferred because real-time systems require fast processing and smooth motion. The computational efficiency of Model A allows it to respond quickly.

Examples include:

- Video games
- Interactive applications
- Virtual reality
- Real-time simulations

For Cinematic Animation:

Model B should be preferred because cinematic applications prioritize realistic and physically accurate motion. Higher computational requirements can be accepted because rendering time is less critical.

Examples include:

- Movies
- High-quality visual effects
- Animated films
- Offline rendering

Conclusion:

The choice depends on the application constraint. **Model A is better for real-time animation** because it is smooth and computationally efficient, while **Model B is better for cinematic animation** because it provides realistic motion despite requiring greater computational power.